The extension–lateral-condition dictionary: self-adjoint completions at infinity versus Feller–Wentzell boundary data of the escape diffusion

Companion note to Theorem 6.1(2)(a)–(b) and Lemma (`lem:feller`)(iii)

Summary. The paper asserts (Thm. 6.1(2)(a)–(b), Lem. `lem:feller`(iii)) that the Feller lateral conditions at the regular boundary of the escape diffusion are "the stochastic mirror" of the $U(1)$ (resp. $U(2)$) family of self-adjoint extensions. This note makes the mirror a theorem and, in doing so, shows it is *not* a bijection: the correct statement is a surjection onto a small, exactly characterized face of the Feller–Wentzell cone, with the unitary phases collapsing to a point fiberwise. The two-sided failure of bijectivity is itself the result: the quantum side carries phase (interference/spectral) data invisible to probability, and the stochastic side carries dissipative and boundary-mass data forbidden to unitarity.

1 Setup and the two parametrizations

Throughout, $\tau\psi = -\frac{1}{2}\psi'' + V\psi$ on the interval $I = (r_-, r_+)$ with $r_+ = +\infty$ an *escaping end*: $V \in C^1$, $V \to -\infty$ eventually monotonically, $|V'|\,|V|^{-3/2}$ bounded (the mechanism model of Thm. 6.1), and classical time of flight $T_+(E) = \int^\infty \mathrm{d}x/\sqrt{2(E-V)} < \infty$, so that r_+ is limit circle for τ [**?**] and a *regular* Feller boundary for the escape diffusion (companion note, Lemma 1(i)). Two cases:

$n = 1$: r_- confining (limit point, no boundary data); deficiency indices $(1, 1)$; extensions $\cong U(1)$.

$n = 2$: $r_- = -\infty$ also escaping with $T_-(E) < \infty$ (e.g. $V = -x^4$); deficiency $(2, 2)$; extensions $\cong U(2)$.

Fix once and for all a real energy E with $V < E$ on a neighborhood of each escaping end—the same E at which the escape diffusion $\mathrm{d}X = \sqrt{2(E - V(X))} \circ (\mathrm{d}t + \varepsilon\,\mathrm{d}W)$ (or its driftless variant, scale $s = \tau_{\mathrm{clock}}$, speed $m = 2\,\mathrm{d}x/v$) is built.

Quantum boundary data (rigorous). For $\psi, \varphi \in D(\tau_{\max})$ the Lagrange bracket $[\bar\psi, \varphi]_x = \bar\psi\varphi' - \bar\psi'\varphi$ has limits at each end, and Green's identity localizes the boundary form at the ends:

$$\langle \tau\psi, \varphi\rangle - \langle \psi, \tau\varphi\rangle = \tfrac{1}{2}\Big([\bar\psi, \varphi]_{r_+} - [\bar\psi, \varphi]_{r_-}\Big) =: i\,\Omega(\psi, \varphi). \tag{1}$$

Ω descends to the $2n$-complex-dimensional boundary space $\mathcal{B} = D(\tau_{\max})/D(\tau_{\min})$, where it is a nondegenerate Hermitian form of signature (n, n); self-adjoint extensions are in bijection with the *maximal* Ω*-neutral* subspaces of $\mathcal{B}$ (complex-symplectic/boundary-triple theory [**?**, **?**, **?**, **?**]). Physically, $\Omega(\psi, \psi)$ is the total *outward probability current* at the ends: with $j(\psi) = \mathrm{Im}(\bar\psi\psi')$ one has $\partial_t|\psi_t|^2 + \partial_x j(\psi_t) = 0$ pointwise in I for *every* extension (all extensions act as the same differential expression), and $\Omega(\psi, \psi) = j(\psi)(r_+) - j(\psi)(r_-)$.

At each escaping end, the solution germs $N_E^{\pm} = \{u : \tau u = Eu \text{ near } r_\pm\}$ form a 2-complex-dimensional space on which the per-end current form

$$J^{r+}(u,w) = \tfrac{1}{2i}\,[\bar{u},w]_{r_+}, \qquad J^{r-}(u,w) = -\tfrac{1}{2i}\,[\bar{u},w]_{r_-} \quad \text{(sign: current } \textit{toward} \text{ the end)}$$

is constant in x, Hermitian, of signature $(1,1)$: choosing real solutions u_1, u_2 with Wronskian 1, the combinations $\psi_\pm = u_1 \pm i u_2$ satisfy $J(\psi_\pm,\psi_\pm) = \pm 1$, $J(\psi_+,\psi_-) = 0$. (Under the mechanism hypotheses the Liouville–Green asymptotics $\psi_\pm \sim v^{-1/2} e^{\pm i\int v\,\mathrm{d}x}$, $v = \sqrt{2(E-V)}$, hold with Olver's error bounds [**?**]; this is used only for the dynamical interpretation, not for the algebra.) Write $u = a\,\psi_+ + b\,\psi_-$ near each escaping end: a is the current-normalized *incident* (toward-the-boundary) amplitude and b the *returned* one, and $J(u,u) = |a|^2 - |b|^2$. Because $T_\pm < \infty$ (limit circle), the trace map $N_E := \bigoplus_i N_E^{(i)} \to \mathcal{B}$ (germs to boundary data) is an isomorphism for every $E \in \mathbb{R}$ [**?**, **?**].

Stochastic boundary data (rigorous; Feller). At the regular boundary r_+ of the escape diffusion, with generator $G = D_m D_s$, the sub-Markov C_0-semigroup extensions of the minimal (absorbed) semigroup are in bijection with *lateral conditions*

$$p_0\,u(r_+) \;-\; p_1\,\frac{\mathrm{d}u}{\mathrm{d}s}(r_+) \;+\; p_2\,(Gu)(r_+) \;=\; \int \big(u(y) - u(r_+)\big)\,\nu(\mathrm{d}y), \tag{2}$$

$p_0, p_1, p_2 \ge 0$, $\nu \ge 0$ a measure on $\bar{I} \setminus \{r_+\}$ (interior and, in the two-ended problem, the other boundary point), not all data zero, modulo positive scaling [**?**, **?**, **?**]. Terminology: p_0 *killing* (absorbing = p_0 alone, i.e. $u(r_+) = 0$, the minimal explosive semigroup; elastic = $p_0, p_1 > 0$), p_1 *reflection*, p_2 *stickiness* (boundary carries a speed-measure atom $p_2\delta_{r_+}$; the process spends positive Lebesgue time at r_+), ν *jump-in* (re-entry, possibly at the other boundary: Feller's "circle-type" passage). Applying (**??**) to $u = \mathbf{1}$ shows: *the extension is conservative iff $p_0 = 0$ at every regular end* (no killing anywhere; ν may be arbitrary). Denote by $\mathcal{W}$ the set of lateral data (one datum per regular end), a cone of infinite dimension (because of ν).

2 The boundary unitary of a self-adjoint extension

Proposition 1 (boundary scattering matrix at fixed energy)**.** *Fix $E \in \mathbb{R}$ as above and let $\tilde{H}$ be any self-adjoint extension, with maximal Ω-neutral boundary subspace $L \subset \mathcal{B}$. Let $\mathcal{S}_L = \{u \in N_E :\ \Gamma u \in L\}$ (germ solutions at energy E admissible for $\tilde{H}$), and let $a = (a_i)$, $b = (b_i) \in \mathbb{C}^n$ be the incident/returned amplitude coordinates at the escaping ends. Then:*

- *(i)* $\dim_{\mathbb{C}} \mathcal{S}_L = n$, *and for $u \in \mathcal{S}_L$, $\sum_i \big(|a_i|^2 - |b_i|^2\big) = 0$* (exact current conservation)*;*
- *(ii) the map $u \mapsto a$ is injective on $\mathcal{S}_L$, hence bijective onto $\mathbb{C}^n$, and there is a unique $S_{\tilde{H}}(E) \in U(n)$ with $b = S_{\tilde{H}}(E)\,a$ on $\mathcal{S}_L$;*
- *(iii) $\tilde{H} \mapsto S_{\tilde{H}}(E)$ is a bijection from the self-adjoint extensions onto $U(n)$. Separated boundary conditions correspond to diagonal S (the torus $U(1)^n \subset U(n)$); for $n = 1$, $S_\theta(E) = e^{i\alpha(\theta,E)}$ recovers the $U(1)$ of Thm. 6.1(2)(a).*

*For comparison, the same construction with "maximal neutral" relaxed to "maximal nonpositive" ($\Omega(u,u) \le 0$ on L, i.e. net outward current ≥ 0: maximal-dissipative extensions, contraction semigroups $e^{-it\tilde{H}}$) yields exactly the closed matrix ball $\|S\| \le 1$ [**?**]; the self-adjoint family is its unitary boundary.*

Proof. (i) The trace map $N_E \to \mathcal{B}$ is an isomorphism (limit circle at each escaping end; the confining end contributes no data), so $\mathcal{S}_L \cong L \cap \mathcal{B} = L$, of dimension n. For $u \in \mathcal{S}_L$: $\Omega(u,u) = 0$ by neutrality of L, and $\Omega(u,u) = \sum_i J^{(i)}(u,u) = \sum_i (|a_i|^2 - |b_i|^2)$ by the constancy of the per-end bracket on solutions.

(ii) If $a = 0$ then (i) gives $\sum |b_i|^2 = 0$, so $u = 0$: injective; by dimension count onto $\mathbb{C}^n$. Define S by $b = Sa$; (i) says $\|Sa\| = \|a\|$ for all a, i.e. $S \in U(n)$.

(iii) $\mathcal{S}_L$ is recovered from S as $\{a\,\psi_+ + Sa\,\psi_-\}$, and $L = \Gamma\mathcal{S}_L$; conversely any maximal neutral subspace of a signature-(n,n) space is the graph of a unitary from the positive to the negative part (if $(0,b') \in L$, neutrality forces $b' = 0$; then the a-projection is bijective and the graph map is isometric). In the (a,b) coordinates at energy E, $\Omega = \mathrm{diag}(+I_n, -I_n)$, so maximal neutral subspaces $\leftrightarrow U(n)$, and by the trace isomorphism these are exactly the extension subspaces. Separated conditions constrain each end's data independently, giving block-diagonal (here 1×1 blocks, i.e. diagonal) S. For the dissipative case replace "isometric graph" by "contractive graph." □

Remark 2 (E-covariance). The extension $\tilde{H}$ is fixed; the frames $\psi_\pm(E)$ rotate with E by an end-local $U(1,1)$ Möbius action, under which unitarity of $S(E)$ is preserved but the moduli $|S_{ij}(E)|$ in general drift with E. Two subfamilies are E-uniform: separated conditions (diagonal S for all E) and the smooth gluing (the paper's periodic identification: the glued clock-circle problem has no defect at the seam, so S is antidiagonal—perfect transmission—at every E). The dictionary below is stated at the mechanism energy E, the same E at which the escape diffusion itself is defined; this matching of reference energies is intrinsic to the construction, not a defect.

3 The dictionary

Definition 3 (flux correspondence)**.** *For a self-adjoint extension $\tilde{H}$ define the lateral datum $\Phi_E(\tilde{H}) \in \mathcal{W}$ by: at each escaping end i,*

$$p_0^{(i)} = 0, \qquad p_2^{(i)} = 0, \qquad p_1^{(i)} = |S_{ii}(E)|^2, \qquad \nu^{(i)} = \sum_{j \neq i} |S_{ij}(E)|^2\, \delta_{r_j},$$

i.e. conservative, instantaneous, boundary-supported, with reflection weight $|S_{ii}|^2$ and passage-to-the-other-end weight $|S_{ij}|^2$. (For $\varepsilon > 0$ the pair (p_1, ν) is to be read projectively: re-entry at local-time rate ν/p_1; in the transport limit $\varepsilon \downarrow 0$ the weights are literal per-arrival branching probabilities. See Remark **??**.*)*

Theorem 4 (extension–lateral-condition dictionary; $n \in \{1,2\}$ escaping ends, $T_i(E) < \infty$)**.** *Let $\mathcal{C} \subset \mathcal{W}$ be the set of lateral data that are simultaneously* (c1) conservative *($p_0^{(i)} = 0$ for all i),* (c2) instantaneous *($p_2^{(i)} = 0$),* (c3) boundary-supported *($\nu^{(i)}$ charges only the other escaping end, not the interior), and* (c4) reciprocal *(for $n = 2$: the passage weights in the two directions are equal, equivalently the reflection weights at the two ends are equal). Then:*

(1) ***(image)*** *Φ_E maps the self-adjoint extensions* onto *$\mathcal{C}$ and* only into *$\mathcal{C}$. The correspondence is not a bijection in either direction:*

(2) ***($n = 1$: total collapse)*** *$\mathcal{C} = \{$pure reflection$\}$ is a single point and Φ_E is the* constant *map: every phase θ corresponds to the reflecting condition $\mathrm{du}/\mathrm{ds}\,(r_+) = 0$. The fiber is the whole $U(1)$: the phase $\alpha(\theta, E)$ fixes the eigenvalue positions (interference data) and has no sub-Markov counterpart.*

(3) ***($n = 2$: a segment)*** *$\mathcal{C} \cong [0,1] \ni R$, the convex family "reflect with weight R, pass through infinity ($r_+ \to r_-$ and $r_- \to r_+$) with weight $1-R$, the same R at both ends." Φ_E is onto: $R = |S_{11}(E)|^2$, realized e.g. by $S = \left(\begin{smallmatrix} \sqrt{R} & i\sqrt{1-R} \\ i\sqrt{1-R} & \sqrt{R} \end{smallmatrix}\right)$; $R = 1$ are the separated completions (e.g. Dirichlet wall), $R = 0$ the wormhole/gluing completions (perfect passage; stochastically Feller's circle-type passage, the clock-circle "darning" of the two boundary points [**?**]). The fibers are the phase tori $\{U \in U(2) : |U_{11}|^2 = R\}$ (dimension ≥ 2; $\dim U(2) = 4$ against $\dim \mathcal{C} = 1$).*

(4) ***(exact exclusions; the asymmetry)*** *No self-adjoint extension corresponds to any lateral datum outside $\mathcal{C}$; each of the four defining constraints is enforced by a distinct exact invariant:*

(c1) killing *($p_0 > 0$: absorbing, elastic) is excluded by unitarity: $\|e^{-it\tilde{H}}\psi\| \equiv \|\psi\|$, while any $p_0 > 0$ datum strictly loses mass ($T < \infty$ makes the boundary accessible from everywhere at every positive time). The absorbing/minimal (explosive) semigroup—the regulator-independent object of Thm. 6.1(2)(c)—is precisely the completion* missing *from the self-adjoint family; it corresponds instead to the maximal* dissipative *completion $S = 0$ (outgoing/radiation condition $b = 0$), and elastic data to strictly contractive $0 < \|S\| < 1$ (Proposition **??**, last part).*

(c2) stickiness *($p_2 > 0$) is excluded because a sticky process holds positive probability* at *the boundary point ($\mathbb{P}(X_t = r_+) > 0$ on a set of times of positive measure; the boundary carries a speed-measure atom, and the semigroup is symmetric only in $L^2(m + p_2\delta_{r_+})$), whereas every self-adjoint extension acts in $L^2(I, \mathrm{d}x)$, whose measure has no boundary atom: $|\psi_t|^2 \mathrm{d}x(\{r_+\}) = 0$ for all t. A quantum counterpart would require enlarging the Hilbert space to $L^2(I) \oplus \mathbb{C}$ (a boundary state)—an extension of a* different *symmetric operator, outside the $U(n)$ family.*

(c3) interior jump-in *($\nu(I^\circ) > 0$) is excluded at the classification level: by (**??**) the boundary form of τ is supported on the ends, so extension domains constrain* only *end data, and $\mathrm{Ext}(H) \cong U(n)$ is exhausted by end-channel unitaries—there is no parameter available to encode an interior re-entry point. (Semiclassically the same: wave packets return through the ends.)*

(c4) asymmetric passage *is excluded by unitarity again: for $U \in U(2)$ the matrix $(|U_{ij}|^2)$ is doubly stochastic, hence $|S_{12}|^2 = |S_{21}|^2$ and $|S_{11}|^2 = |S_{22}|^2$. Independent lateral data at the two ends form a 2-parameter family (R_+, R_-); the image is exactly its diagonal.*

(5) ***(dynamical content; semiclassical)*** *For a WKB wave packet of mean energy E launched toward end i: it reaches the boundary after the classical time of flight ($\approx T_i(E)$, the $\varepsilon \downarrow 0$ explosion time of the escape diffusion, companion Lemma 1(ii)), and the fraction $|S_{ji}(E)|^2$ of its probability re-emerges from end j after the finite round trip—the paper's revival numerics (Dirichlet wall: return from the right; periodic gluing: return from the left) are the two extreme points $R = 1, 0$. This matches the $\varepsilon \downarrow 0$ escape diffusion with branching lateral datum $\Phi_E(\tilde{H})$, for which the boundary behavior per arrival is literally "reflect w.p. R / pass w.p. $1 - R$."*

Proof. (1)–(3): By Proposition **??**, $\tilde{H} \mapsto S(E)$ is a bijection onto $U(n)$, and Φ_E factors through it. For $n = 1$, $|S| = 1$ forces reflection weight 1: the constant map; the fiber is $U(1)$. (Equivalently, without Proposition **??**: any separated real boundary condition $\cos\theta\,\Gamma_1 u + \sin\theta\,\Gamma_2 u = 0$ is real, so its admissible germ line is spanned by a *real* solution $u = a\psi_+ + \bar{a}\psi_-$, whence $|b/a| = 1$: total

reflection for every θ.) For $n = 2$, double stochasticity of $(|S_{ij}|^2)$ (row and column sums 1) gives $|S_{11}|^2 = |S_{22}|^2 =: R$ and $|S_{12}|^2 = |S_{21}|^2 = 1 - R$, so the image lies in the diagonal family of (3), which is contained in $\mathcal{C}$ by construction of Φ_E. Ontoness: the displayed matrix is unitary for every $R \in [0, 1]$. Fibers: fixing the moduli of a 2×2 unitary leaves the phases, a torus of dimension ≥ 2.

(4): (c1) Unitarity versus strict mass loss: if $p_0 > 0$, start the process at any $x \in I^\circ$; since r_+ is regular and accessible ($\Sigma < \infty$; companion Lemma 1), $\mathbb{P}_x(\text{hit } r_+ \leq t) > 0$ for every $t > 0$, and each boundary visit kills a positive fraction of local time mass, so $\mathbb{E}_x[\mathbf{1}_{\{\text{alive}\}}] < 1$ for $t > 0$; no mass-preserving evolution can correspond to it under any pairing that matches total probability. The dissipative statements are Proposition **??**: maximal nonpositive boundary subspaces are contractive graphs $\|S\| \leq 1$; $S = 0$ is the condition $b = 0$ (no returned amplitude: every admissible germ is purely outgoing), the quantum radiation condition, whose semigroup loses exactly the arriving flux, as does the absorbing/minimal semigroup; $0 < \|S\| < 1$ returns the fraction $\|Sa\|^2/\|a\|^2 < 1$, as does the elastic condition (per-arrival kill fraction $1 - |S|^2$ in the transport limit).

(c2) For a sticky datum, $\mathbb{E}\int_0^t \mathbf{1}_{\{X_s = r_+\}} \mathrm{d}s > 0$ after the first boundary hit (speed atom $p_2\delta_{r_+}$; [**?**], §3.7), so $\mathbb{P}(X_s = r_+) > 0$ on a time set of positive measure, while the quantum state space $L^2(I, \mathrm{d}x)$ assigns the boundary zero mass at all times. Any pairing matching the measure class of the state space excludes it.

(c3) Green's identity (**??**): the form Ω vanishes on compactly supported interior functions, i.e. the self-adjointness constraint is a condition on end data alone; the classification $\mathrm{Ext}(H) \leftrightarrow$ maximal neutral subspaces of the *end* boundary space is exhaustive, so no extension encodes an interior channel; Φ_E's jump measures are supported on $\{r_-, r_+\}$ by construction, and no refinement of Φ_E through boundary data can do otherwise.

(c4) is the double-stochasticity computation above.

(5) Proof sketch (the only non-exact leg). Under the mechanism hypotheses the Liouville–Green solutions carry Olver-type error bounds ($\int |v^{-1/2}(v^{-1/2})''| \, \mathrm{d}x < \infty$ follows from $|V'||V|^{-3/2}$ bounded and $V \to -\infty$ monotonically), so $\psi_\pm = v^{-1/2} e^{\pm i \int v}(1 + o(1))$ with controlled remainders. A packet $\int \chi(\lambda) u_\lambda \, e^{-i\lambda t} \mathrm{d}\lambda$ built on admissible solutions ($b(\lambda) = S(\lambda)a(\lambda)$) has, by stationary phase, incident front arriving at the end at the classical time and returned fronts of weight $|S_{ji}|^2$ emerging at end j, with corrections $O(\text{window}^{-1}) + o(1)_{\text{WKB}}$; the discrete spectrum (spacing π/T_{cross}, Thm. 6.1(2)(b)) does not obstruct because the packet width covers many levels. On the stochastic side, $\varepsilon \downarrow 0$ reduces the escape diffusion to the transport flow $\dot{x} = v$ (companion Lemma 1(ii): $\zeta \to T$ a.s.), for which the branching lateral datum acts per arrival with probabilities $R, 1 - R$. The two evolutions then agree at the level of arrival times and returned/passed mass fractions. This leg is asymptotic by nature (the exact statements are (1)–(4)). □

Remark 5 (rate normalization at $\varepsilon > 0$). At $\varepsilon > 0$ a regular boundary is visited on a local-time set, so "reflect w.p. R per bounce" is ill-posed; the correct ν-vs-p_1 normalization is an excursion-theoretic convention (jump at local-time rate $\propto (1 - R)/R$), and only the $\varepsilon \downarrow 0$ limit makes the weights literal per-arrival probabilities. We therefore claim the quantitative identification of R only in that limit; the class-level statements (which face of the cone, which ends linked, reciprocity) are exact for every $\varepsilon > 0$.

Remark 6 (why the phases must collapse). The collapse of the $U(1)$ fibers is structural, not an artifact of the map chosen. On the escaping end $v \geq v_{\min} > 0$, so the quantum phase $\int^\infty v \, \mathrm{d}x = \infty$ diverges while the clock $\int^\infty \mathrm{d}x/v = T < \infty$ converges: the quantum boundary basis $\psi_\pm$ winds *infinitely often* approaching the end (solutions have infinitely many zeros accumulating at $+\infty$),

whereas the diffusion's boundary basis—the G-harmonic functions 1 and s—is monotone with finite boundary limits. No linear identification of the two 2-dimensional boundary spaces can be canonical (there is no preferred "zeroth winding"), and the only basis-independent quantities transferable to the probabilistic side are the current moduli $|S_{ij}|^2$. The phase is precisely the datum the $|\cdot|^2$ functor forgets.

Remark 7 (what would see the phases). Two natural extensions of the stochastic side, both outside Feller's lateral-condition cone for the escape diffusion in $L^2(m)$, are the plausible receptacles for the lost data, and we record them as *conjectural* dictionary entries: (a) sticky data $p_2 > 0$ correspond to self-adjoint operators on $L^2(I) \oplus \mathbb{C}$ (a boundary degree of freedom; Wentzell-type quantum boundary conditions), not to any extension of H in $L^2(I)$; (b) the Wigner delay $\partial\alpha/\partial E$ of a fixed extension is the natural phase shadow of boundary delay, but it can be negative and is not constant-per-visit, so it does *not* match any sub-Markov sticky parameter—consistent with (c2) being an exact exclusion rather than a near miss.

4 The correspondence table

Quantum completion (at E)	**Flux data $S(E)$**	**Feller–Wentzell lateral datum**	**status**
any self-adjoint ext., one end (every phase θ)	$e^{i\alpha}$, $\lvert S\rvert = 1$	reflecting: $\mathrm{d}u/\mathrm{d}s(r_+) = 0$	proved (class); timing WKB
separated completions, two ends (θ_+, θ_-)	diagonal, $R = 1$	independent reflection at each end	proved
smooth gluing ("periodic through ∞")	antidiagonal, $R = 0$, all E	passage $r_+ \leftrightarrow r_-$ (circle-type jump; clock-circle darning)	proved (class)
generic coupled $U \in U(2)$	$0 < R < 1$	reflect wt. R + jump to other end wt. $1 - R$, same at both ends	proved (class); R generically E-dependent; rate conv. Rem. **??**
relative phases $\arg S_{ij}$	invisible	— (fibers of Φ_E; eigenvalue positions)	proved
maximal dissipative, radiation cond. $b = 0$	$S = 0$	absorbing $u(r_+) = 0$: minimal explosive semigroup	proved (flux); *not* self-adjoint
strictly contractive boundary data	$0 < \lVert S\rVert < 1$	elastic ($p_0, p_1 > 0$: partial kill)	proved (flux); *not* self-adjoint
none in $L^2(I, \mathrm{d}x)$ (needs $L^2 \oplus \mathbb{C}$)	—	sticky ($p_2 > 0$)	exclusion proved; enlarged-space entry conjectural
none (no interior channels in $\mathrm{Ext}(H)$)	—	jump-in to interior ($\nu(I^\circ) > 0$)	exclusion proved
none (double stochasticity)	—	asymmetric passage $R_+ \neq R_-$	exclusion proved

Verdict

PARTIAL-WITH-CHARACTERIZED-IMAGE; the class-level dictionary is PROVED.
The correspondence between self-adjoint extensions and Feller–Wentzell lateral conditions is *not*

a bijection and should not be presented as one. Precisely (Theorem **??**): the flux functor Φ_E is a surjection of the $U(n)$ extension family onto the face $\mathcal{C} = \{\text{conservative} \wedge \text{instantaneous} \wedge \text{boundary-supported} \wedge \text{reciprocal}\}$ of the Feller–Wentzell cone—a *point* (pure reflection) for one escaping end, a *segment* $R \in [0,1]$ (reflection vs. reciprocal passage through infinity) for two—with phase-torus fibers; the paper's wormhole completion is exactly the passage datum $R = 0$. Absorbing and elastic data are matched not by self-adjoint but by maximal-*dissipative* completions ($\|S\| \leq 1$ matrix ball, sub-unitary $\leftrightarrow$ sub-Markov, unitary boundary $\leftrightarrow$ conservative face); sticky and interior-jump data have no counterpart in $L^2(I, \mathrm{d}x)$ at all. All exclusions and the image characterization are exact; only the dynamical (wave-packet arrival/revival) interpretation is semiclassical (PROOF-SKETCH-SOLID), and the $\varepsilon > 0$ jump-rate normalization is conventional. For the paper: Thm. 6.1(2)(a)–(b) and Lem. `lem:feller`(iii) are correct as worded ("stochastic mirror," "parametrize"), provided "mirror" is not sharpened to a bijection; the safe sharp citation is: *the conservative instantaneous boundary-supported reciprocal lateral conditions are exactly the flux shadows of the self-adjoint family, the phases being spectral data with no stochastic counterpart.*

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